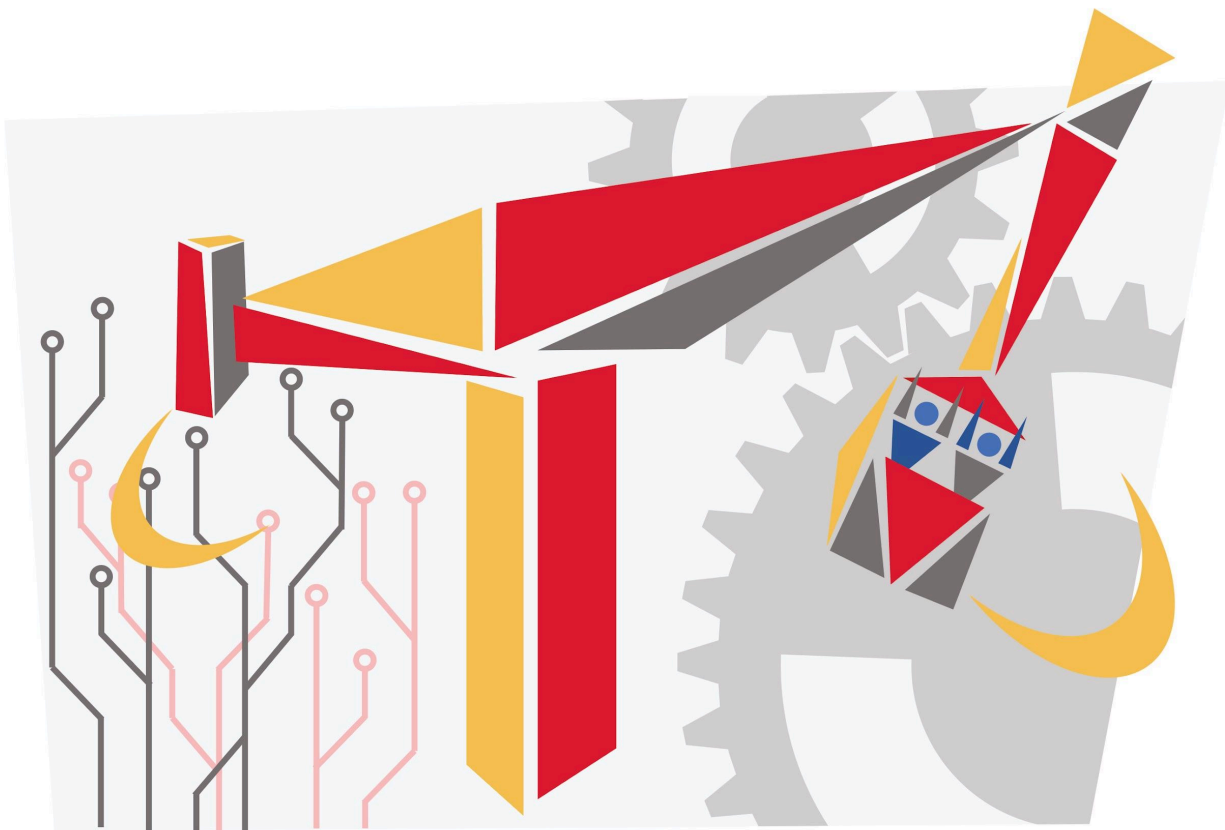


Ride Engineering Competition



Rule Book 2026

Ride Engineering Competition

Ride Engineering Competition Inc. is not responsible for any physical or financial harm as a result of activities described in official documentation or travel to and from official events, and it is not responsible for confidentiality of any information obtained from participating entities and persons.

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Introduction and Synopsis

Rationale for the creation of this competition:

Students seeking a career in the Themed Entertainment industry have the opportunity to demonstrate their themed experience and attraction design skills in several existing competitions. These competitions often focus on Industrial Design concepts, or short term design decisions, but do not allow for a full lifecycle Systems Engineering Process. Collected data shows that most University Theme Park clubs include high concentrations of mechanical, civil, and electrical engineering students who have not participated in competitions outside the Attractions Industry. Project Based Learning is a vital part of higher engineering education. This competition seeks to provide a Multidisciplinary Project Based education with specific amusement industry practices to students that participate.

Overview:

The Ride Engineering Competition offers students a chance to demonstrate real-world engineering practices within the industry focus of ride engineering and design. Student teams will have 6 months to design, plan, and manufacture an attraction for small candy riders based on a prompt that includes a plot of land and desired rider experience. Teams will receive points for three major deliverables: a Systems Engineering Report that details the engineering decisions and actions throughout the project, a Functional Physical Model of the attraction, and a team score for Video Theory of Operation and Outreach. Teams will also receive point deductions for the relative cost of their ride compared to their competition.

The Competition culminates in an event featuring a 6-hour window in which all Rides must run continuously. During this time Teams will Service their rides whenever needed. Judges will award points based on the ride's uptime, theme, service plan, and Site Acceptance Testing.

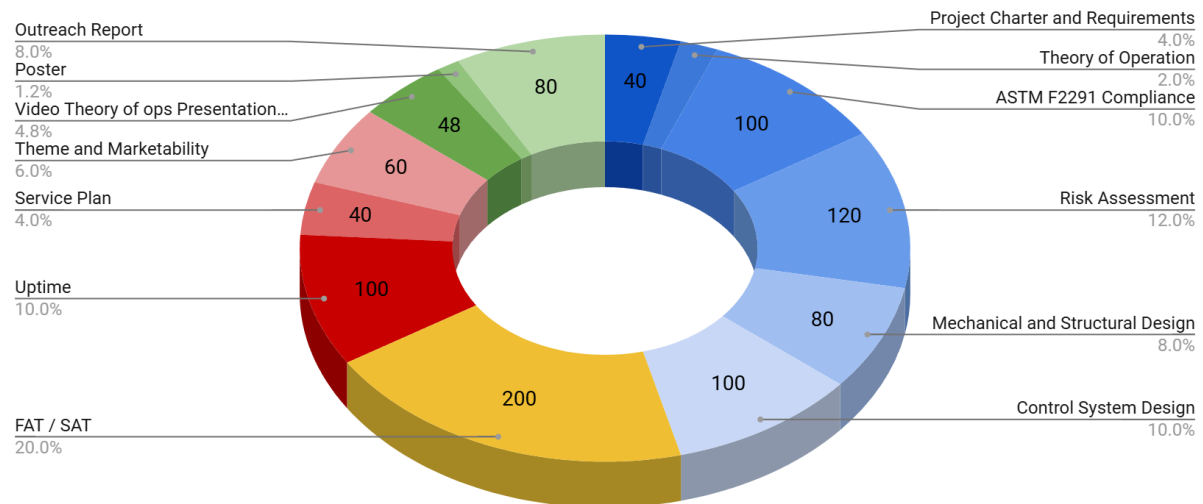
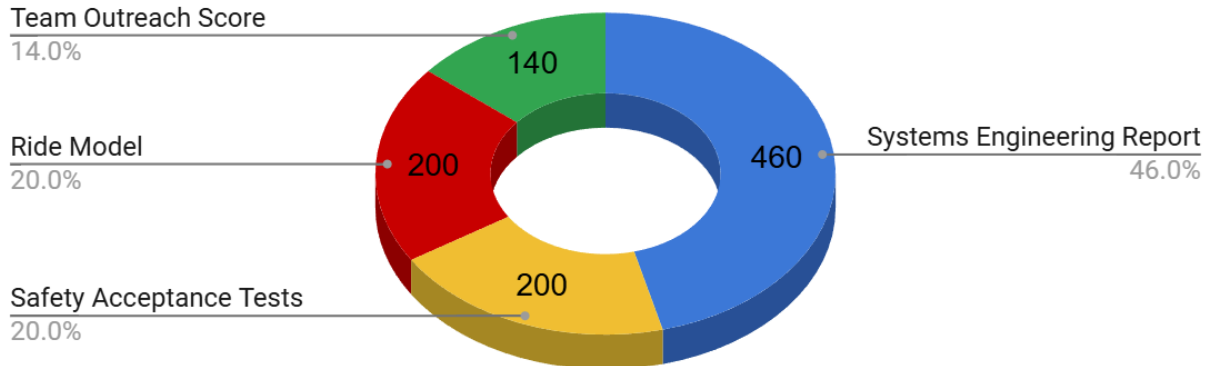
Timeline

See full Schedule of Deliverables at the end of the rule book

- **September 27 2025:** Registration Opens and Prompt is Announced
- **November 8 2025:** Registration Closes
- **4 weeks after registration:** Project Charter due
- **January 17 2026:** Preliminary Design Review
- **February 28 2026:** Final Design Review,
Theory of Operation, ASTM F2291 Compliance due
- **March 29 2026:** Final Hand-in and Bag & Tag
- **April 4 2026:** Poster, Video Theory of Operations due
- **April 11 2026:** Competition Event

Scoring

Scoring Overview



There are 1000 points up for grabs, however only 750 points are expected for teams that perform at the level of a professional ride engineering team. Additional points can be earned in each section by going above and beyond and impressing our judges. Scoring above 500 points will be a great achievement.

For more details see the detailed Rubric at the end of the Rule Book.

Competition Events

The Competition will be held on-site on April 11th, 2026

The Day of Operation

Teams shall arrive with their ride disassembled within the Portable Envelope. An REC official shall verify that the ride is within the portable envelope and ready the team to begin assembly at the Opening of the park. Once this period is done, an official shall announce the Opening of the Park. At this time, all teams will start assembling their Rides, efficiently bringing them into automatic operation. The Park will be open for 6 hours. All Rides should continue automatic operation for this entire span of time. At a pre-scheduled time, teams will halt their operation and disassemble their Rides back into the Portable Envelope. An REC official shall verify that the ride is within the portable envelope and allow the team to re-assemble their ride and return it to automatic operation. Teams are recommended to have 1 member present with their ride at all times, although this is not a requirement. If a judge or official notices a ride seemingly malfunctioning with no team members present, they will attempt to halt the ride and notify the Team Leader of the Downtime and observations.

Downtime

If an attraction malfunctions, Teams must service the attraction and restore it to an operating condition. It is the responsibility of the Team to prepare for these repairs. Teams may voluntarily suspend operation of their ride to service it.

All Servicing actions will be documented with a Downtime Form detailing the circumstances that caused the downtime and the actions the Team took to resume operation. A timer will be counting the uptime of the ride. The timer shall be suspended whenever a human enters the ride area.

Physical Ride Judging

During the Operation Day, groups of Judges will be meeting with teams to see their Rides in action. Judges will question the teams on Marketability, Theming, and Service Plans, and will conduct Site Acceptance Tests on each ride.

Teams will receive notice 30 minutes before groups of judges will visit to discuss these sections. Time is limited for these visits. Ensure that your team is prepared with any additional resources or documentation. Judges will award points in these categories based solely on this interaction, and it may be the only interaction these specific judges have with your team or report.

Team Rules

- Eligible Competitors: Higher education students enrolled during the entirety of the competition season.
- Majors: This is a multidisciplinary competition. Having team members from multiple fields of study, especially Mechanical Engineering AND Electrical/Controls Engineering is strongly advised.
- Teams must consist of a minimum of three higher education students and one advisor. (4-10 recommended)
- The advisor must be faculty or staff of an academic institution that represents at least one of the team members (in case of members from multiple institutions) If you are a student organization, you most likely have an advisor already.
- The required involvement of the advisor is to be a contact and liaison to the academic institution. Advisors are encouraged to mentor the teams but must not significantly contribute to the design or fabrication of competition entries.
- Graduate Students and Registered Students in Internships as part of their degree program may also participate.
- There is no maximum student member limit on a single team.
- Teams may consist of members from multiple institutions.
- One institution may have multiple teams.
- Advisors may advise more than one team.
- Teams may not work in direct collaboration with other teams, however all are encouraged to practice “coopertition” by communicating and assisting the community of teams with specific issues for the improvement of all teams. Teams at the same institution must work independently.

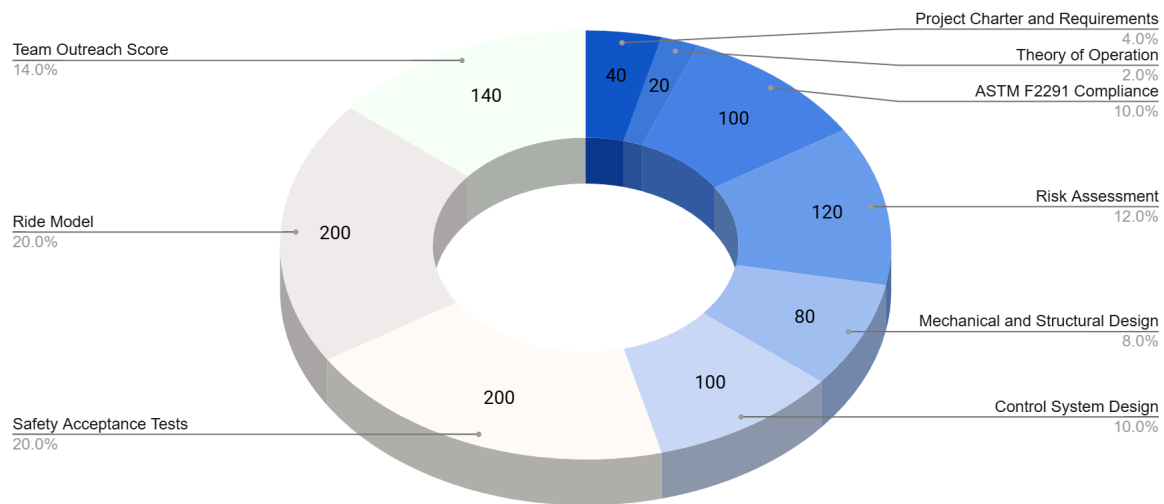
For any questions about team formation or conduct, contact the coordinators.

Judging Guidance

Each section will be scored by a separate judge. This judge will have experience in that subject matter, but they will not be an expert on your ride. Judges will be instructed to view all Theory of Operation documents to receive a quick introduction to the ride system and rider experience. Each section shall contain all of the required information and context needed to demonstrate the points earned including quoted references to other sections of the competition. Energy spent focusing on concise and clear communication for an uninformed audience is historically received well by our judges.

Systems Engineering Report (460 pts.)

The purpose of the Systems Engineering Report is to promote a Systems Engineering approach to the challenge. The Report accounts for 46% of a team's overall score and is designed to emulate many of the real engineering tasks conducted by engineers in the attractions industry. The designs referred to in the report detail the ride that the team creates, and will reflect the materials, components, dimensions, and forces used in the physical ride. The required contents of the Report can be broken down as follows:



- **Project Charter and Requirements (40 pts.)**

The Project Charter is a vital part of an engineering task. The purpose of a Project Charter is to provide a framework to define the scope and desired outcome of the project. This document is to be created before any substantial engineering work is performed and must include:

- Problem Statement
 - Define the ultimate goal of the project, and the means by which that goal will be achieved. This should be written without any specific design decisions.
- Background Context and State-of-Art research
 - Gather information that may inform or inspire your team during your design process.
- Stakeholders
 - Identify all parties internal and external who will affect the project in any way. This includes the REC Planning Committee.
- Initial Resources
 - Identify the people, availability, skills, tools, funds, and opportunities you have available to your team. This will help you identify needed resources during the project.
- Scope of Work
 - Identify the deliverables, timeline, and limits of work for the project. A Gantt Chart is highly recommended for creating your project timeline.
- Design Criteria and Desired Outcomes
 - Requirements from competition documents and team constraints shall be noted in as much detail as possible. Create metrics that can be used to evaluate the project during the season.

The Project Charter should be created before any significant development is done on the project and agreed upon by all members of the team. This tool can be used throughout the project by creating updated versions at key milestones of the project, such as Design Reviews.

The Project Charter submitted 4 weeks after Team Registration will be scored for points.

- **Theory of Operation (20 pts.)**

The Theory of Operation is an overview of what the final ride is, and what it does. This section shall include an organized introduction to the systems and subsystems of the ride. The details of how those systems function shall be covered in the Mechanical and Structural Design, and Control System Design sections.

Outline each function and provide descriptions, images, or diagrams to clearly and concisely communicate what your ride is at a systems level. All judges will be directed to the Theory of Operations as the introduction to your ride.

The Theory of Operation shall be due along with your proof of Final Design Review.

Specific points for this section are low, but previous results have shown that teams who effectively introduced their ride scored significantly higher in subsequent categories.

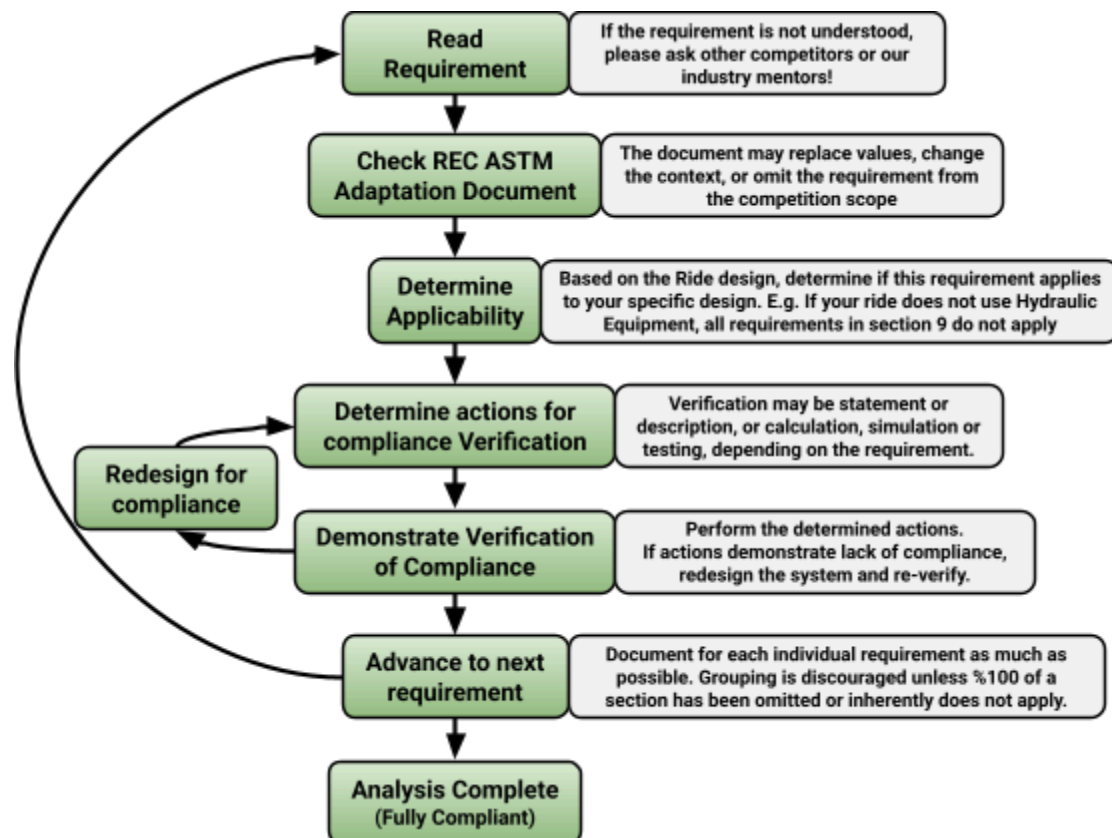
- **ASTM F24 Compliance Analysis (100 pts.)**

Teams shall demonstrate that their attractions comply with ASTM F24 standard F2291-25 (or later) pertaining to the Attraction Designer / Manufacturer. Teams will use the ASTM F2291 Standard along with the Ride Engineering Competition ASTM Adaptation Document providing guidance vital to compliance within the context of this competition. The REC ASTM Adaptation Document includes information on which sections do not apply within the context of the competition, as well as figures to be used in place of those in F2291. Teams are to use the Adaptation document as rationale for their compliance analysis.

E.g. Section 5.3.1.4 does not apply to this ride because the REC ASTM Adaptation Document states that section 5.3.1.4 does not apply to the Ride Engineering Competition.

Teams may demonstrate compliance with additional standards, as industry professionals are required to do, for additional points.

Teams are encouraged to take a systematic approach to identifying applicability and compliance to every requirement in F2291 with the following process:



Ride Engineering Competition

The majority of points in this section are awarded for thoroughly assessing each individual requirement and determining if it applies to your ride. This can be efficiently documented in a list or table.

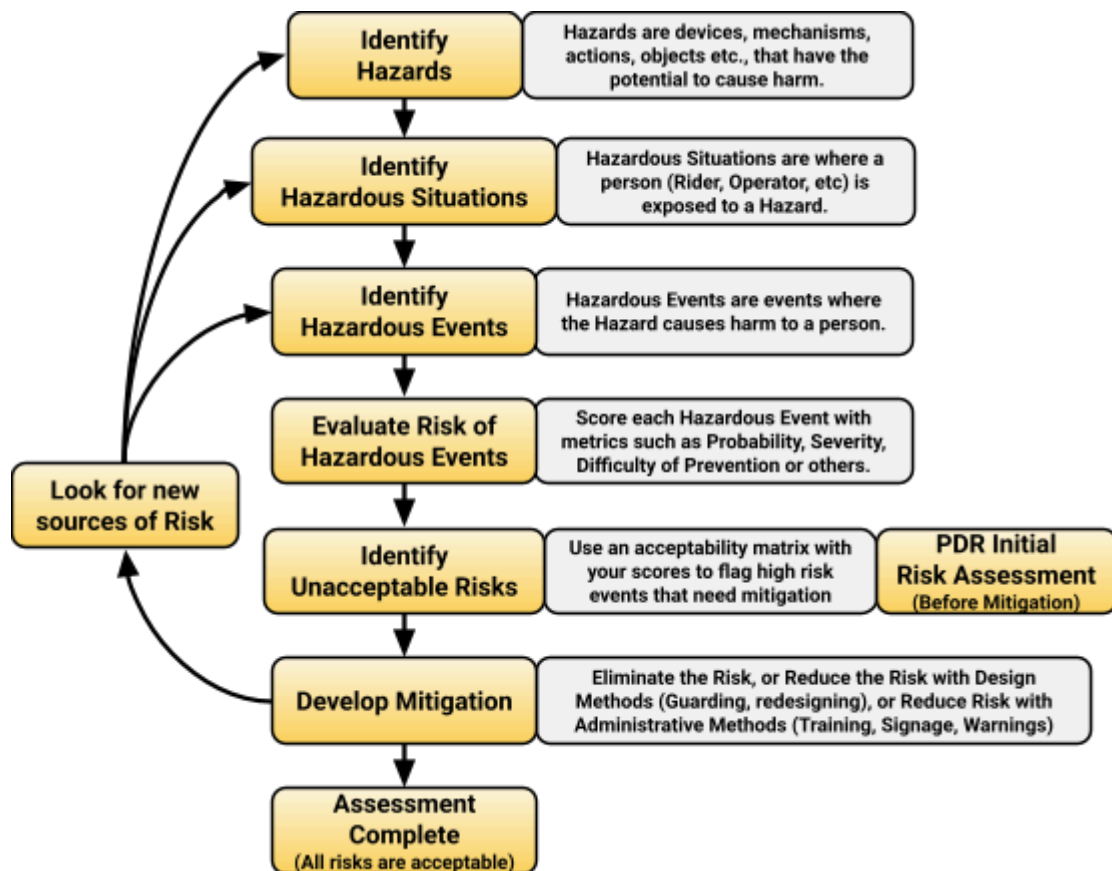
Verification of your compliance is accomplished through references to the features of your ride outlined in your Theory of Operation, Calculations and Analysis outlined in your Design sections, or Tests conducted in your Factory Acceptance Testing. If specific Tests or Analyses have not yet been conducted at the time of submittal, each shall be described in this section.

The ASTM F24 Compliance Analysis shall be due along with your proof of Final Design Review.

- **Risk Assessment (120 pts.)**

This document outlines the potential risks on People (Riders, Operators, etc.) presented by the Ride. All reasonably foreseeable hazards and risks should be identified and evaluated using criteria such as Severity and Probability, and possibly other metrics such as Exposure or Difficulty of Prevention. The combination of these valuations shall generate a risk value. Hazardous Events that score above an acceptable value shall be mitigated via Elimination, Design Methods, or Administrative Methods. Teams shall demonstrate a thorough understanding and assessment of potential risks.

The Initial Risk Assessment shall be prepared for the Preliminary Design Review (PDR) after Identifying and scoring Risks based on the Ride Concept. Additional Risks and Mitigation methods shall be included in the final Risk Assessment section of a team's Systems Engineering Report.



Within the safety industry, Risk Assessments are built to address harm to people and do not address financial, social, reliability, or other negative outcomes. Risk Assessments begin with a Hazard List and identify the safety features and functions needed to mitigate or eliminate each hazardous event. As a result of this perspective, the Risk Assessment can be conducted before details of the design have been finalized. The inverse of a Risk Assessment is a Failure Modes Effects Analysis (FMEA) which starts with detailed components and identifies the outcomes of their failure. Both are valuable tools, but the Risk Assessment is the most efficient way to achieve an acceptable level of safety.

Note: What is harm to an inanimate rider? As engineers it is up to the teams to determine what they constitute as "harm" to their riders. There is no "correct" interpretation. The goal is to have sound and thorough justification for all assumptions.

- **Mechanical and Structural Design (60 pts.)**

The Theory of Operations, Risk Assessment, and ASTM Compliance will designate structures and mechanical functions that your ride will need, along with the safety requirements of those functions. The Mechanical and Structural Design section is the place to detail and engineer the components that make up that structure. This shall include descriptions, visuals and drawings, and calculations including but not limited to hand calcs, Finite Element Analysis, and simulations. The methods for these calculations may be affected by ASTM requirements.

Safety critical calculations shall be verified in Factory Acceptance Testing.

A full Drawings package shall be created for all parts and systems within the ride. A viewer should be able to completely manufacture the functional ride based solely on these documents. Judges will award points for Engineering Drawings, Electrical Schematics, and Additional Documents.

Engineering Drawings of all components shall adhere to a consistent drawing standard (ASME Y14.100 recommended).

- **Control System Design (80 pts.)**

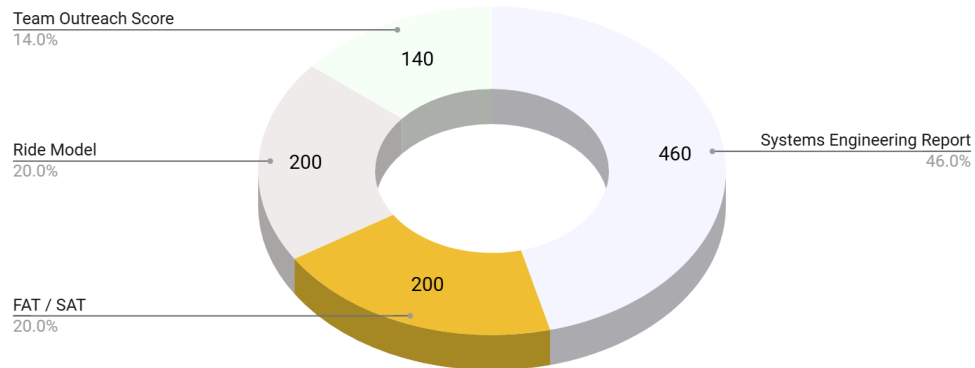
Functional Safety Management is a process of ensuring safety by enacting functions that mitigate risk of hazardous events. Functional Safety Management mindset is a guide that uses a Risk Assessment to inform the design of all safety related mechanical and electrical controls systems.

The Theory of Operations, Risk Assessment, and ASTM Compliance will designate safety-related digital, electrical, and mechanical controls that are needed to mitigate risk and ensure an acceptable level of safety. The Control System Design section is the place to detail and engineer the components and processes that make up that Safety-Related Control System. This shall include descriptions of functions, visuals and schematics, and calculations of safety parameters including but not limited to hand calcs and simulations.

Safety critical calculations and functions shall be verified in Factory Acceptance Testing.

Electrical Schematics shall document the connections between components and specify electrical characteristics of each connection (Voltage, Protocol, etc). Programmed Code shall also be included.

Acceptance Testing (200 pts.)



Earn points through FATs, Defend points through SATs

Teams shall manufacture their functional ride model and validate their Safety systems through a series of Factory Acceptance Tests (FATs). The documentation, design, and successful completion of these tests will earn up to 200 points.

The exact functions and components to be tested will be derived from the Mechanical and Structural Design, and Control System Design Sections, with additional considerations from the ASTM Compliance and Risk Assessment sections as necessary.

Before the competition event, teams shall submit a **Site Acceptance Testing Plan** outlining a maximum of 3 tests that will be conducted on the Competition Day. During the Competition Event, teams will be given a 10 minute window to conduct their chosen SATs for the Judges . The clear successful completion of these tests and compliance with their written test plan shall allow the team to retain up to the number of points earned by their Factory Acceptance Tests.

E.g. A team that earns 150 points in their FAT, and scores 80% in their SAT, will receive a final score in this section of 120 points.

Teams shall label all FAT procedures with a test number. Missing test numbers will result in a point penalty.

Examples of test number styles:

Test 47: Normal Operation

Test 6.1.4: Normal Operation

Test 14a: Normal Operation

Teams shall identify the completion status of any tests (Completed, Failed, Untested)

Teams shall classify all tests with one of the following 3 labels.

- **Component FAT**
 - Tests for individual components (single parts, subassemblies, material tests, etc.) of the attraction
- **Full System FAT**
 - Tests that require the full assembly and operational capabilities of the attraction.
- **SAT**
 - Maximum of 3 tests that shall be presented to the judges to demonstrate the proper function of the attraction and its safety critical features.

Disclaimer: It is expected that all tests are appropriately identified and conducted exactly as presented in the documentation. (Ex: A benchtop component test for a drive motor shall not be classified as a Full System FAT, nor shall it follow any procedure outside the one explicitly written by the team for the drive motor).

The Systems Engineering Report as a whole outlines a single process in which Teams shall use their Risk Assessment to determine the Mechanical and Electrical Systems needed to meet the Theory of Operation and ASTM Requirements, provide Rationale and Calculations for the specific implementation of each function, Detail the test needed to verify the proper functioning of the system, and Provide evidence that the functional ride model has successfully performed that test.

Design Reviews

Design reviews act as key components to the engineering process where the team presents their work at a milestone and receives feedback from other members of their team and external knowledgeable guests such as outside students, professors, or professionals. Teams that do not have connections to industry professionals are encouraged to contact the REC Planning Committee. Engineering and Safety are universal languages, and any individual with similar expertise from any industry can provide valuable feedback and guidance during these reviews.

During the Competition Season, Teams are required to perform 2 design reviews.

The Preliminary Design Review shall cover the initial concept for the ride and an initial risk assessment for the ride. The review may cover additional content at the discretion of the team.

The Final Design Review shall cover the Theory of Operation, ASTM Compliance, and design progress of the ride, empowering the team to move with confidence into the final engineering and manufacturing of the ride itself.

Teams are required to send validation to the REC that these reviews were conducted. This could be in the form of a summary, meeting minutes, a photo, the presentation, or a recording. Reviews can be conducted at any time before the deadlines.

The REC will review the progress of concepts at these reviews and may contact individual teams to assist with ensuring compliance with the rules and prompt.

Bag & Tag

Rides will be fully complete and functional, passing Factory Acceptance Tests before the hand-in deadline for all reports two weeks before the final competition event. Rides may not receive additional work or modification after that date. To ensure that Rides are not modified during that week, all Rides must be placed in a Bag or Bags, sealed with a one-way ratcheting tie, Tagged with the Team and Date and Time of seal, as provided or as described by the REC Committee. A photo of the sealed Ride must be submitted alongside the hand-in documents to clearly show that the Ride has been sealed in the Bag. All components of the Ride must be sealed in the Bag including Electronics and Operator Interfaces.

Additional temporary materials and objects may be placed in the Bag to support safe transportation of the Ride. The Ride should be disassembled into the Portable Envelope before being placed in the Bag. The Bag or Bags may occupy any shape or size.

If necessary for the safe transportation of the ride (For example disassembly of ride components to fit into a suitcase or other container) Teams may re-Bag their ride by providing clear photos or video and documentation of the reason for the re-Bag and the before and after states showing the re-sealed ride. Teams must obtain approval from the REC Committee before opening their bag to avoid a penalty, as this year's challenge is designed to eliminate this issue.

Officials at the Competition Event will review the sealed Ride for tampering before authorizing teams to unseal their ride. Teams will be allowed to unseal their ride and ensure it is within the portable envelope. Officials will verify the portable envelope before teams may start assembling their ride and prepare it for the day of operation.

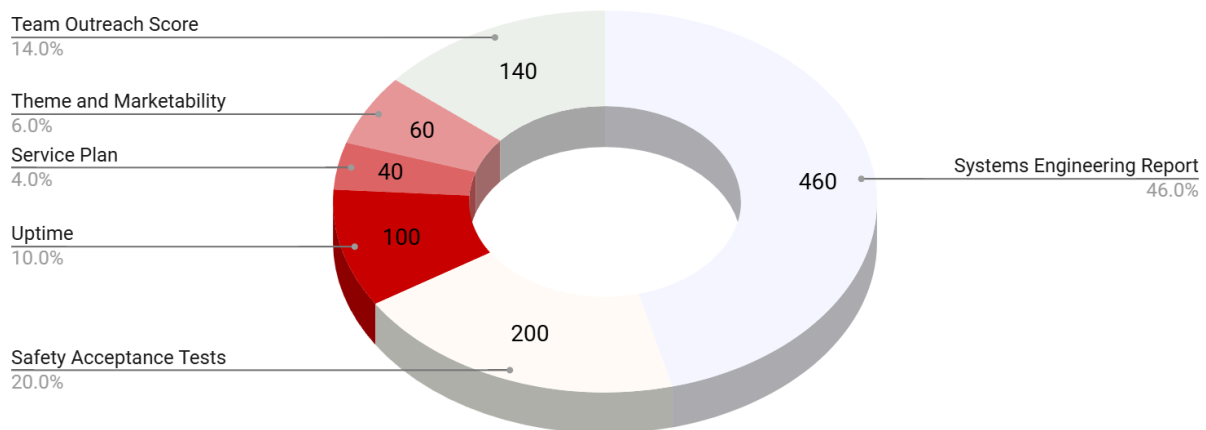
Teams are advised to expect potential damage during transportation and bring the appropriate spare materials and tools needed to perform needed repairs on the competition day.

Physical Ride (200 pts.)

Teams will demonstrate their designs by manufacturing a functional attraction for wrapped candy Riders. The Ride shall perform the ride experience and theme of the attraction with the functional mechanical and control systems detailed in the Systems Engineering Report and the Factory Acceptance Testing.

Note on rider scale: This is not a model ride. The riders of this attraction are candies as described in the Rule book. These riders are not stand-ins for human riders. These candy riders are the real riders of this real attraction. The accelerations and forces used to engineer the ride shall be based on the actual accelerations and forces experienced by the candy riders.

The Ride will be given points during the competition day based on the following categories:



- **Uptime (100 pts.)**

This score takes into account the Ride's ability to continuously run throughout the Park's Operating Hours.

Points will be rewarded for the percentage of time during the operation period that a ride is running. Teams can earn back points lost through effective service execution and may earn additional points than the number lost for exceptional performance.

No physical contact with the ride itself should be needed while in operation. Any physical contact with the ride will trigger downtime and must be reported. Safety measures shall be built into the control system, including an Emergency Stop to take immediate action in the occasion a crisis emerges.

If a ride receives catastrophic damage and becomes Standing But Not Operating at any point throughout the Operating Hours, it will not receive any additional points in this section for the remainder of the event. Teams should prepare for this and have spare parts on hand in the emergence of such an event.

- **Service Plan (40 pts.)**

Teams shall prepare a Service Plan including a Failure Modes Effects Analysis to prepare for the likely failures that may occur and the necessary service procedures required to address them. The Service Plan should include Failure Modes, Inspection Criteria, and Service Procedures.

Teams will receive points for efficiently, quickly, and safely servicing their rides while striving for maximum Uptime.

- **Theme and Marketability (60 pts.)**

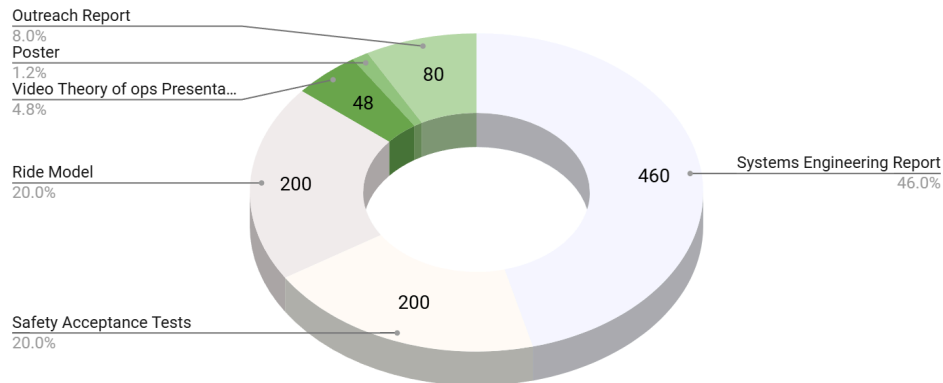
An effective ride theme portrays a story to guests and immerses them in an atypical environment. Teams will use theme elements such as colors, props, geometry, and any type of storytelling necessary to captivate guests and enrich the ride experience. Rides and theming should be realistic and true to scale with the rider.

An effective theme builds on the ride experience of the ride system and can be effectively marketed to potential guests. Teams shall demonstrate to judges how the theme is portrayed to riders experiencing the ride, and how it is marketed to attract new riders. Marketing elements include, but are not limited to, the ride's name, color scheme, logo, tagline, and more.

A ride's theme should reflect the ride experience, thrill level, and target audience. Teams will receive points for creating a theme that is congruent with the ride system and thrill level of their ride.

Ride experiences should satisfy the prompt request and be observingly fun and adventurous to an average guest. Points will be awarded for rides that create visual interest, by way of thrill, intrigue, or kineticism. by offering for example heights, spins, and/or exciting and new sensations to the rider. Rides may be any kind of experience and are allowed to be similar in form or function to existing attractions.

Team Outreach (140 pts.)



- **Video Theory of Operations Presentation (48 pts.)**

Teams shall create a video no longer than 5 minutes that summarizes the ride system in the style of the Theory of Operations. Videos will receive points for effective technical communication and creativity.

- **Poster (12 pts.)**

Teams will also create a 18" x 24" Poster, 300 dpi or less, giving an overview of their ride, process, safety features and any other details they would like to feature. These posters will be printed by the REC Committee and displayed at the competition event for a public audience alongside the ride itself.

- **Outreach Report (80 pts.)**

Every team has a responsibility to give back to their communities in the form of STEAM communication and education. Teams will submit an outreach paper detailing events and programs where they have enriched their community with education of Ride Engineering concepts and STEAM (Science, Technology, Engineering, Arts, and Math) education. The paper should include information such as what events and programs the team has created/contributed to, who the event/program impacted, lasting effects of the event/program, etc. Only events that occurred within the period of time between the previous competition and this year's event are eligible to receive points. For ongoing events please include relevant context and history.

The Bid (-100 pts.)

Each Team must present a Proposal Bid with their ride, evaluated and itemized at the value of all materials and manufacturing labor used to construct the Ride. Team bids will be compared and teams will be deducted points based on the value of their Bid relative to their competition. The Team with the highest value Bid will receive a deduction of 100 pts. The Team with the lowest value bid will receive no deduction. Remaining teams will receive deductions tiered linearly within that range based on their relative value rank. (e.g. If 5 teams are competing, deductions will be of 0 pts., 25 pts., 50 pts., 75 pts., and 100 pts.)

All materials included in the final assembly of the ride must be included in the bid at the value at which they were purchased. If this is unknown, a primary source price listing (from manufacturer or retailer) may be used as reference.

Manufacturing time performed by the team must be added to the Bid at a rate of \$20.00 per hour. This only means time spent permanently altering parts or creating new parts. This cost is a direct trade-off to purchasing that same part. Assembly of created parts does not need to be included in the Bid. Engineering and Design Labor is not included in the Bid.

3D Printed parts are discouraged and will have a manufacturing cost of \$1.00 per gram.

Note: 3D printing adds an extremely high cost to the Bid Value (\$1.00/g compared to approx \$0.02/g for PLA filament), this is a deliberate action by the REC Planning Committee. 3D Printing can offer an unfair advantage to teams with access to 3D printing machines. Hobby 3D printing can create poor habits relative to the desired learning outcomes of the REC in terms of use of Engineering Drawings for Manufacturing, Manufacturing Quality Assurance, and Designing for sourced Commercial-Off-The-Shelf (COTS) parts. 3D printing is allowed and teams are encouraged to use it for small highly functional or aesthetic components, but seek alternative manufacturing methods for the majority of their ride.

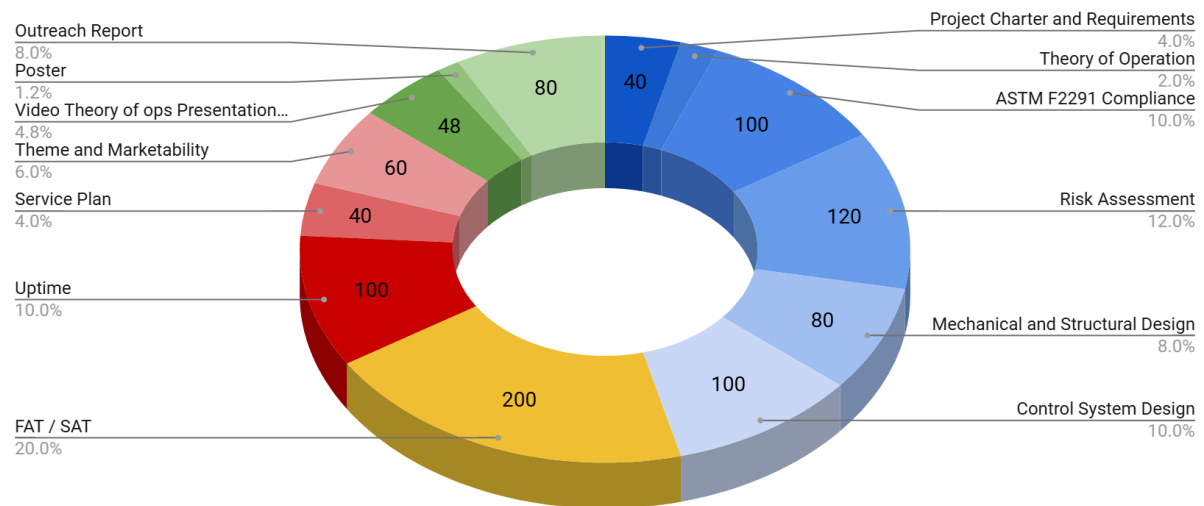
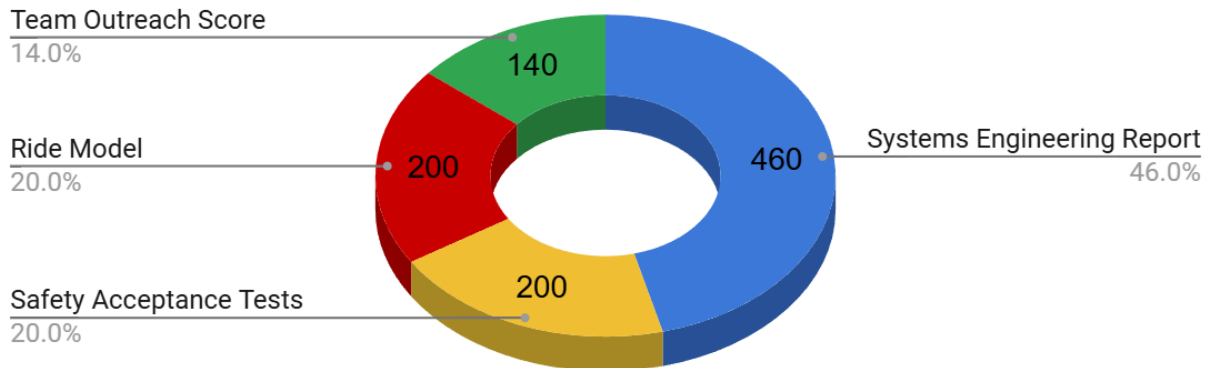
Spare parts used in servicing the ride during competition are not included in the Bid.

Bid documents are encouraged to be itemized tables with unit prices that sum to the final value.

Team bids determined to be inaccurate or lacking citation will be penalized in the rankings appropriately to the perceived inaccuracy by the judges.

Bids in previous years have ranged from \$150 to over \$7,000.

Scoring Overview



Summary of Deliverables

Name	Section	Score	Filename	Due Date
Project Charter	Systems Engineering Report	40	Teamname 2026 PC	4 weeks after registration
Proof of Preliminary Design Review on Initial Risk Assessment and Concept	REC Review	-	Teamname 2026 PDR	January 17, 2026
Proof of Final Design Review	REC Review	-	Teamname 2026 FDR	February 28, 2026
Theory of Operation	Systems Engineering Report	20	Teamname 2026 TOPs	February 28, 2026
ASTM F2291 Compliance Analysis	Systems Engineering Report	100	Teamname 2026 ASTM	February 28, 2026
Risk Assessment	Systems Engineering Report	120	Teamname 2026 RA	March 29, 2026
Mechanical and Structural Design	Systems Engineering Report	80	Teamname 2026 Drawings	March 29, 2026
Control System Design	Systems Engineering Report	100	Teamname 2026 Mech FAT	March 29, 2026
Factory Acceptance Tests	FAT	200	Teamname 2026 CS FAT	March 29, 2026
Bag & Tag Proof	REC Review	-	Teamname 2026 Bag Tag	March 29, 2026
Outreach Report	Team Score	80	Teamname 2026 Outreach	March 29, 2026

Ride Engineering Competition

Bid	Ride Model	(0, -100)	Teamname 2026 Bid	April 4, 2026
Service Plan	Ride Model	40	Teamname 2026 Service Plan	April 4, 2026
Poster	Marketability	12	Teamname 2026 Poster	April 4, 2026
Video Theory of Operations Presentation	Team Score	48	Teamname 2026 Video TOps	April 4, 2026
Functional Ride at Event	Ride Model	-	-	April 11, 2026

Submit Deliverables by Midnight (Local time for each team) on the due date on the REC Website:

<https://www.rideengineeringcompetition.org/>

Failure to submit Deliverables by the deadline may result in point deductions.

Presentations and Events

Name	Section	Score
Site Acceptance Tests	Ride Model	Up to 200
Exposition of Theming and Marketability	Ride Model	60

Full Scoring Rubric			
Item	Points Available	Additional Points for Excellence	Total
Systems Engineering Report	345	+115	460
Project Charter	30	+10	40
Problem Statement In one sentence, what is the ultimate goal of this project?	3		
State-of-Art Research What similar products exist? How are they similar to this project? How will this project be different? What can we learn from what currently exists?	4	+2	
Stakeholders Who are the parties involved with the project, both internally and externally? What must be communicated between these parties and how and when does this communication occur.	2		
Initial Resources Teams shall identify what resources are currently possessed by the team and what resources need to be obtained before completion of the project. This includes personnel, information, workspaces, tools, materials, funding, etc.	3		

		Scope of Work Teams shall identify what work is being performed and what party is responsible for that work. This shall include identification of Deliverables, and estimated schedule of work (e.g. Via a GANTT chart)	8	+3	
		Design Criteria Teams shall identify important criteria that must be met to satisfy the Problem Statement. These criteria shall be directly measurable with estimated values included.	10	+5	
		Theory or Operation	15	+5	20
		Ride Overview Teams shall describe the desired rider experience and the systems that make up the ride.	5	+2	
		Subsystem introduction Teams shall give a description of each subsystem in the ride and its functions.	10	+3	

	ASTM F24 Compliance Analysis	75	+25	100
	Identification of Applicable Requirements Teams shall demonstrate a thorough assessment of all requirements within ASTM F2291-25b, and for each declare if that requirement is applicable to their ride. Teams shall justify with appropriate context each declaration.	50	+15	
	Verification of Compliance For each passage that is deemed applicable to the ride system, teams shall show that the compliance is met through analysis, testing, calculation, and/or appropriate pieces of documentation."	25	+10	
	Risk Assessment	90	+30	120
	Thorough identification of Potential Hazards by all components to all people Are all reasonably foreseeable hazards identified? Are all aspects of the ride properly analyzed?	40	+15	
	Valuation of risk based on multiple risk factors and high-risk threshold What criteria are used to determine level of risk? Are values appropriate to the corresponding hazard? Are high-risk situations accurately identified?	15	+5	
	Actions taken for high-risk outcomes What actions are taken to reduce or eliminate the risk? How effective are these actions? Are new hazards introduced by this action assessed?	35	+10	

Mechanical and Structural Design		60	+20	80
	Design Description and Rationale Mechanical and Structural components referenced from the Theory of Operation and Risk Assessment shall have details defined including geometry, materials, functional parameters, etc. This shall include descriptions, visuals and drawings.	20	+5	
	Calculations Teams shall create a complete calculation package to outline all safety related components and structures including but not limited to hand calcs, Finite Element Analysis, and simulations	25	+5	
	Mechanical Drawing Package A complete drawing packing shall be presented for all components with full details ready for manufacture.	15	+10	

	Control System Design		75	+25	100
	Design Description and Rationale Safety-Related Control Functions referenced from the Theory of Operation and Risk Assessment shall have details defined including components and functional parameters. This shall include descriptions, visuals and Schematics.		25	+5	
	Calculations Teams shall create a complete calculation package to outline all safety-related functions including but not limited to hand calcs, Finite Element Analysis, and simulations		30	+10	
	Electrical Schematics and Code Package Electrical Schematics shall document the connections between components and specify electrical characteristics of each connection (Voltage, Protocol, etc). Programmed Code shall also be included.		20	+10	
				Sub - Total	460

Item	Points Available	Additional Points for Excellence	Total
Acceptance Testing	150	+50	200
Factory Acceptance Tests	150	+50	200
Overall Testing Plan <i>What did you choose to test?</i> Teams shall generate a comprehensive series of Component and Full system Tests to verify Mechanical and Structural Calculations and the Safety-Related Control System. The tests shall be numbered and clearly organized.	50	+20	
Testing Procedures <i>Did you write high quality tests?</i> Each Test shall be defined in terms of the system or function being tested, the conditions of the test, the parameter of a successful result, and the means of recording the result.	50	+20	
Results <i>Did you complete and learn from your tests?</i> Teams shall conduct the Tests described, show the results of the tests, and describe any and all actions taken because of the passing/failing of the test. Failed tests are eligible for full points. Tests shall be conducted exactly as presented in the documentation.	30	+10	

		Site Acceptance Testing Plan Characterization <i>Do these chosen tests validate the safe function of the full ride?</i> Teams shall define 3 tests to be conducted at the final event in a separate document. Points shall be awarded for the degree to which these 3 tests effectively characterize the proper function of the attraction and its safety critical features.	20		
	Site Acceptance Tests		Up to 100% of FAT Points		
		Procedure <i>Can you follow your written test plan?</i> Teams shall conduct the tests they presented in their Site Acceptance Testing Plan . Judges shall evaluate the team's adherence to the written SOP, ensuring all steps are followed as they are written.	50%		
		Results <i>Was your test successful?</i> Teams shall complete their tests. Judges will evaluate the results of the tests to ensure they were successful.	50%		
				Sub - Total	200

Item	Points Available	Additional Points for Excellence	Total
Functional Ride	175	+25	200
Uptime	100	0	100
Uptime Percentage of the Operational period that the Ride is functioning without intervention.	100	0	
Incident Penalty Major incidents will yield additional penalties based on the severity of the incident on riders. (Minimum score is 0/100)	Max -20 per rider for ejection		
Service Plan	30	+10+	40
Failure Modes Analysis Teams shall document the effects of each foreseeable component failure in the ride system.	15	+5	
Inspection Criteria Teams shall document key indications that failures have occurred or are likely to occur.	5	+2	
Service Procedures Teams shall detail the tasks needed to safely and effectively react to failure modes and return the ride to safe operation.	10	+3	

Ride Engineering Competition

		Service Execution Teams will receive points for efficiently, quickly, and safely servicing their rides while striving for maximum Uptime.		+ Amount determined by judges	
	Theme and Marketability		45	+15	60
		Implementation Judges shall award points for the overall effectiveness and clarity of the theme as built on the ride.	10	+3	
		Story Teams shall demonstrate how the theme is communicated to riders experiencing the ride.	15	+6	
		Marketing Teams shall demonstrate how the ride and theme can be communicated to potential new riders.	10	+3	
		Innovation Teams shall communicate the qualities of their ride concept that demonstrate new advances in ride design, both for local guests and the industry as a whole.	10	+3	

Ride Engineering Competition

	Bidding	-100	0	-100
	Bid ranking Each Team must present with their ride a Proposal Bid evaluated at the value of all materials and manufacturing labor used to construct the Ride. Team bids will be compared and teams will be deducted points based on the value of their Bid relative to their competition. The Team with the highest value Bid will receive a deduction of 100 pts. The Team with the lowest value bid will receive no deduction. Remaining teams will receive deductions tiered linearly within that range based on their relative value rank.	-100		
			Sub - Total	200

Item	Points Available	Additional Points for Excellence	Total
Team Outreach Points	120	+40	160
Video Theory of Operations Presentation	33	+15	48
Technical Communication Teams shall create a video no longer than 5 minutes that summarizes the ride system in the style of the Theory of Operations. The video shall clearly introduce the system to a technical but uninformed audience.	22	+10	
Creativity Teams shall find effective ways to communicate their technical content within the format of a short video.	11	+5	
Poster	12	0	12
Informational Poster 18" x 24" Poster, 300 dpi or less, giving an overview of their ride, process, safety features and any other details they would like to feature. Presenting any poster rewards all points (all or nothing)	12		

	Outreach		60	+20	80
	Outreach Report		60	+20	
	Teams shall create a report outlining outreach events in which team members participated including details of the lesson plan and how it was communicated, demographics of the audience reached and number of people reached. Events that promote STEAM education (Science, Technology, Engineering, Art, and Math) and careers in the Attractions industry are encouraged. Teams that demonstrate a strong presence and positive impact on their community will earn additional points.				
				Sub - Total	140

Summary			Points Available	Additional Points for Excellence	Total
Systems Engineering Report			345	+115	460
Acceptance Tests			150	+50	200
Functional Ride			175	+25	200
Team Outreach			105	+35	140
			775	+225	
				Total	1000

Safety Requirements

When competing at REC, **YOU** are responsible for your own safety. This includes:

- Using power tools correctly.
- Using hand tools correctly.
- Donning on personal protective equipment (PPE).
- Ensuring tools brought to REC meet the list of allowed items.
- Ensuring work areas are secure, if repairs are needed.
- Submitting a Safety Data Sheet (SDS) for any epoxies or chemicals that may be brought onto site.
- Donning on extra PPE, if needed (facial masks for sanding).

Allowed	Allowed in Designated Zone	Not Allowed
Scissors Wrenches Screwdrivers Sand paper (<i>when done by hand</i>) Pre-approved CA Glue and Epoxy Calipers Multimeters Concrete (<i>Fully contained, No pouring on day</i>) (<i>Silica Respiratory Hazard</i>)	Drills (<i>No gloves, Sacrificial surface</i>) Hot Glue Gun (<i>Heat Resistant Gloves recommended, not required, mitigated by first Aid</i>) Sanding (<i>Respiratory hazard</i>) Cutting (<i>Cut Resistant glove on non-cutting hand</i>) Soldering Irons (<i>Lead free solder</i>) Heat Guns	Pocket Knives Power Cutting Tools Blades above 1" (<i>Except scissors</i>) Open Flames

These requirements are similar to what Environmental Health and Safety (EHS) departments at theme parks will ask. Not only will these requirements help keep you and those around you safe during the competition, but this will give you a step up in knowing safety requirements that will come up, when you work in this industry. Though you are responsible for your own safety, ensure that you are keeping an eye out for your fellow competitors and if you see something, say something.

If there are any safety related issues that happen during the event, please flag down one of the REC team members and they will be sure to address it accordingly.